BASICS OF GenAI

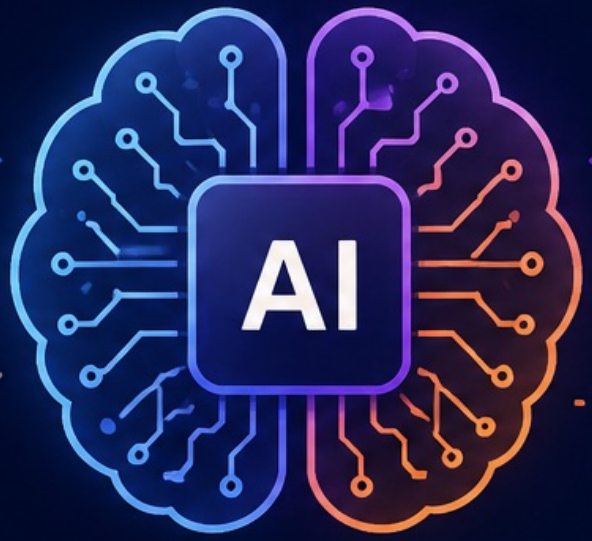
Understand. Create. Innovate.



TEXT GENERATION



IMAGE GENERATION



CODE GENERATION



IDEAS TO IMPACT





UNDERSTAND



CREATE



INNOVATE



EMPOWER

Lecture: Prompt Engineering 2

Agenda

1. Opik demo
2. Chain of Thought
3. React
4. How to write prompts for RAG

Opik (Comet product)

↳ product for llmops

↳ Track: prompts, usage (token), cost, error

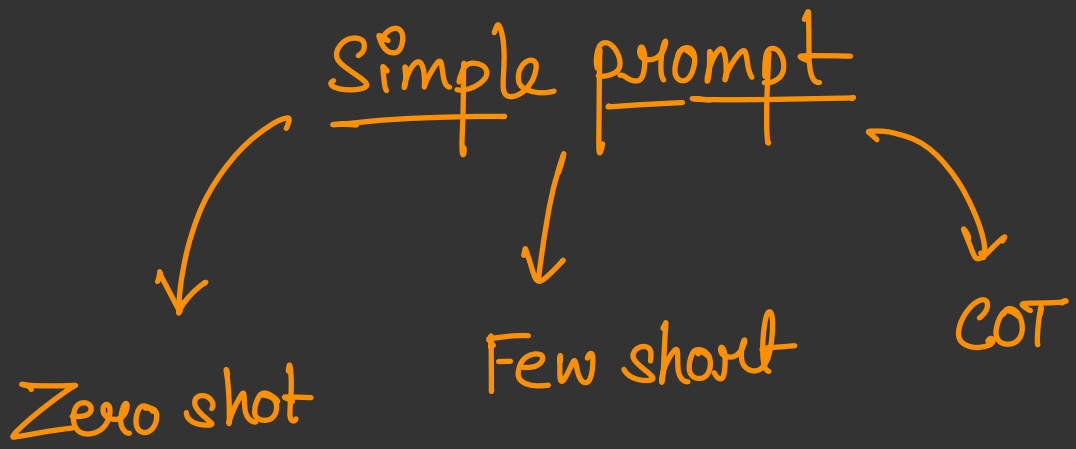
(Dashboard)

prompt Engg

1) Zero Shot → No example, direct ask.

2) Few Shot → Few example with direct ask.

3) COT → prompt with step by step breakdown


Simple prompt
Zero shot
Few shorut
COT

Chain of Thought

↳ Mimic how humans think.

Instead of direct answer → step by step approach.

Useful: Reasoning Task

Core idea: Improve performance

1) Decompose Complexity
2) Reduce errors
3) Improves quality & accuracy

1) Zero shot COT

↳ Lets think step by step / Think carefully before answering

2) Few shot COT

↳ Same as above + examples.

3) Structured COT

↳ "First Analyze X, get the result. then analyze Y"

ReAct

Reasoning + Action

"COT + External Tool"

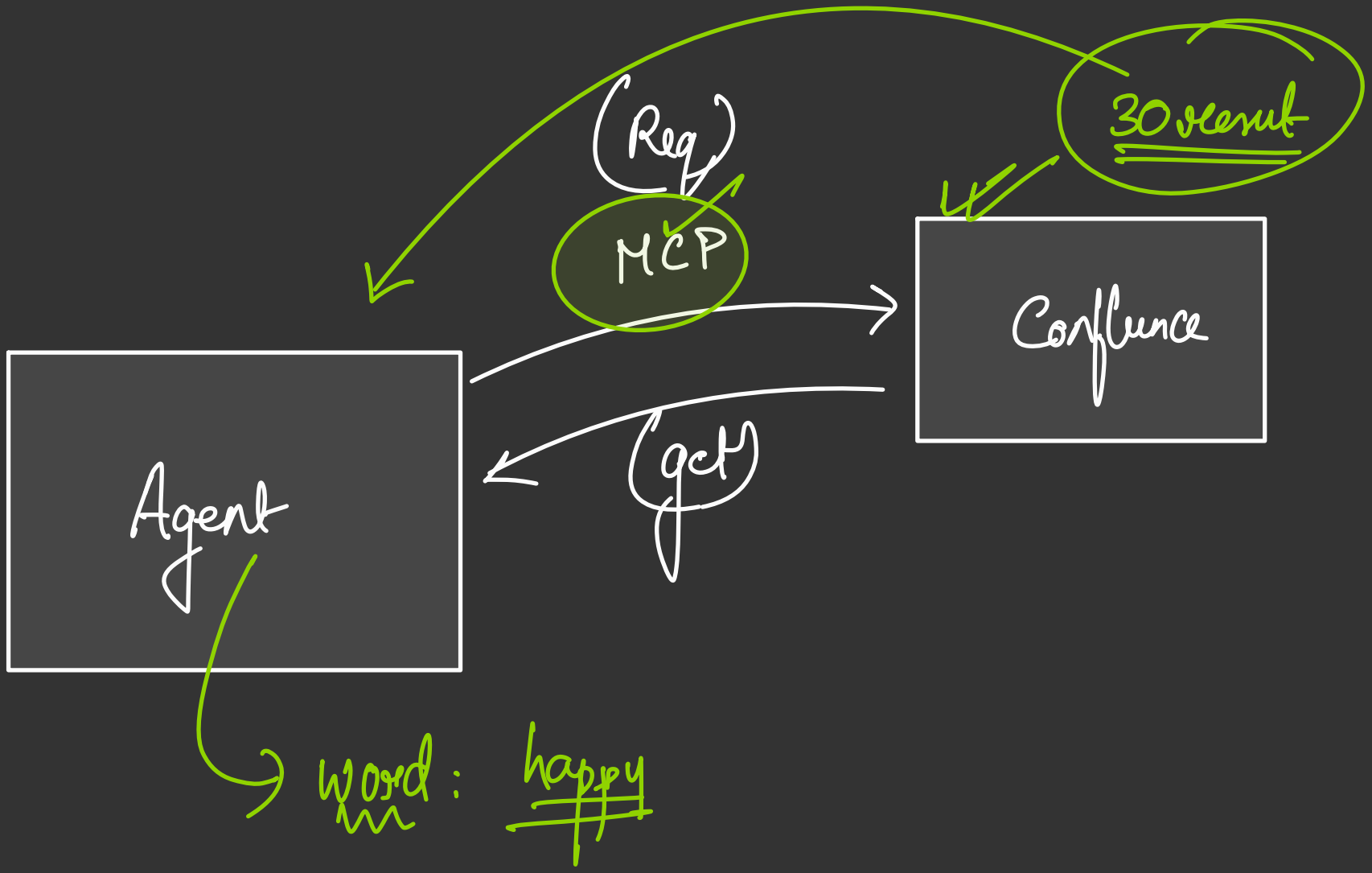
→ Calendar, DB, API, websearch

think → act → observe → repeat

What is the weather in Tokyo?

pydanticAI

↳ Steve


Req
MCP
30 result
Confluence
Agent
get
word : happy

Agent

Thing

1) goal
2) access to tools
3) Memory
4) Ability to take action

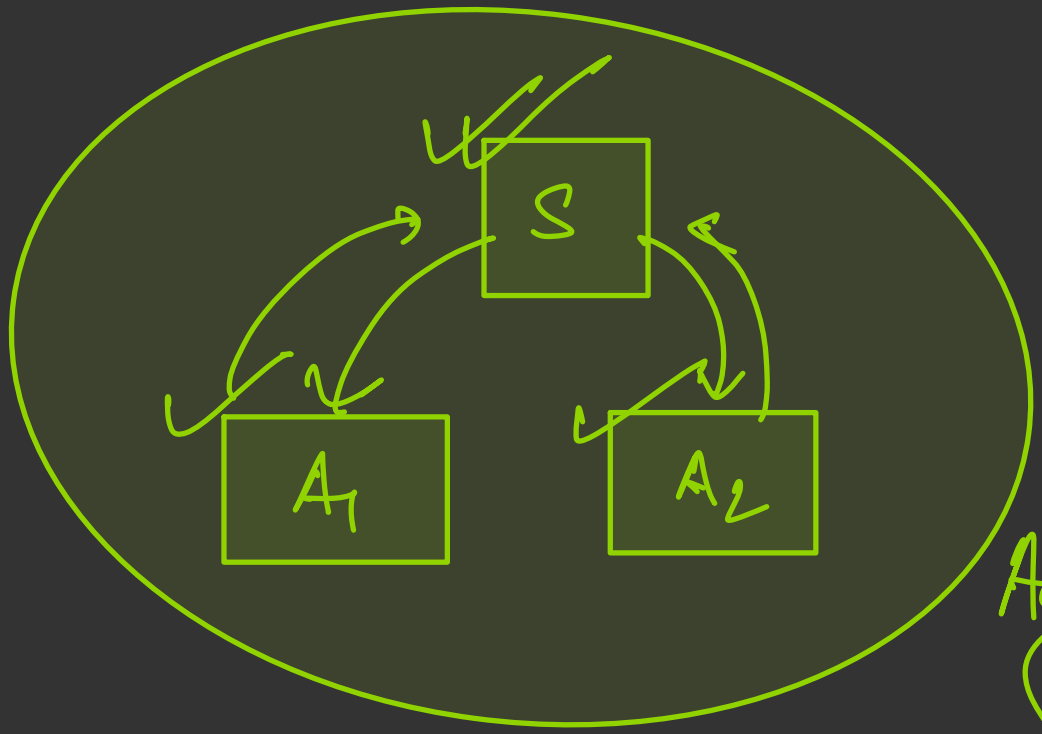
Agentic AI

The behaviour

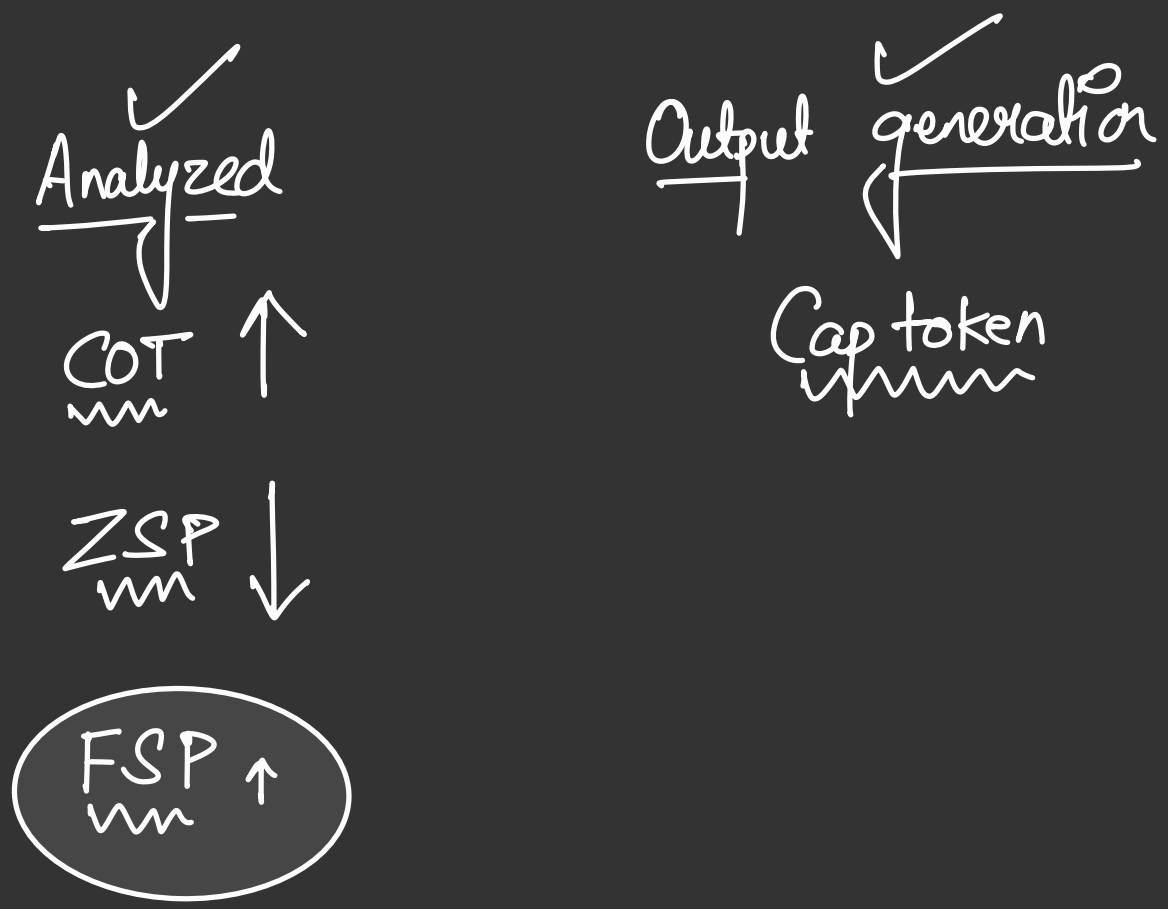
1) Planning multi-step process
2) Use tools autonomously
3) Adapt to thing when needed
4) Rollback if thing go wrong

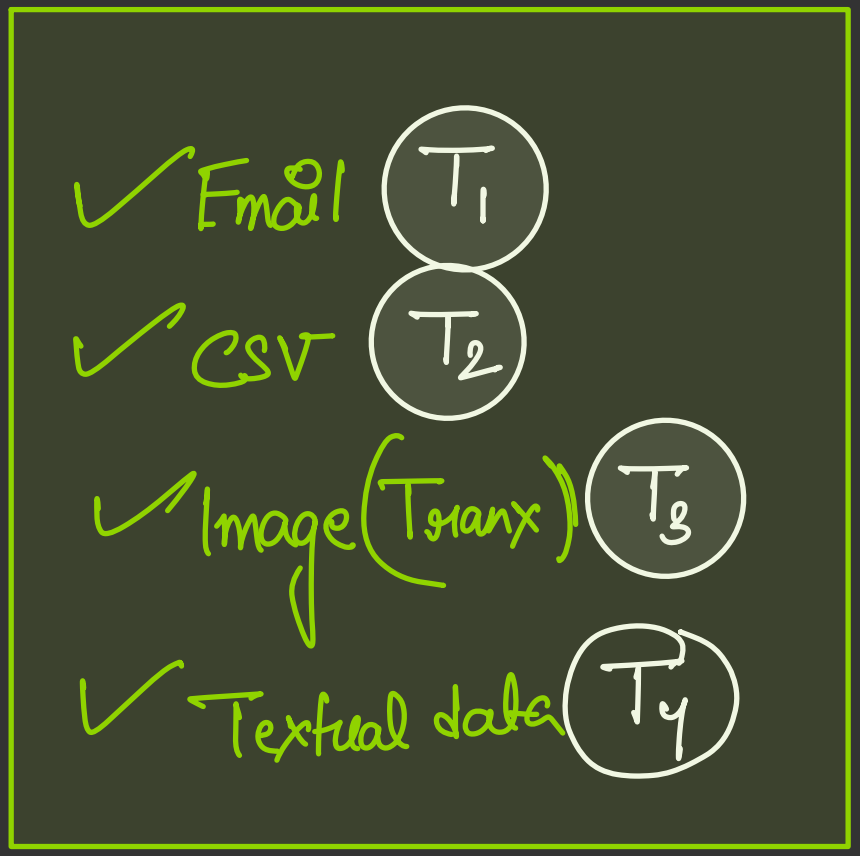
Agent → Employee

Agentic AI → the concept of Emp working together towards an obj



Agentic AI

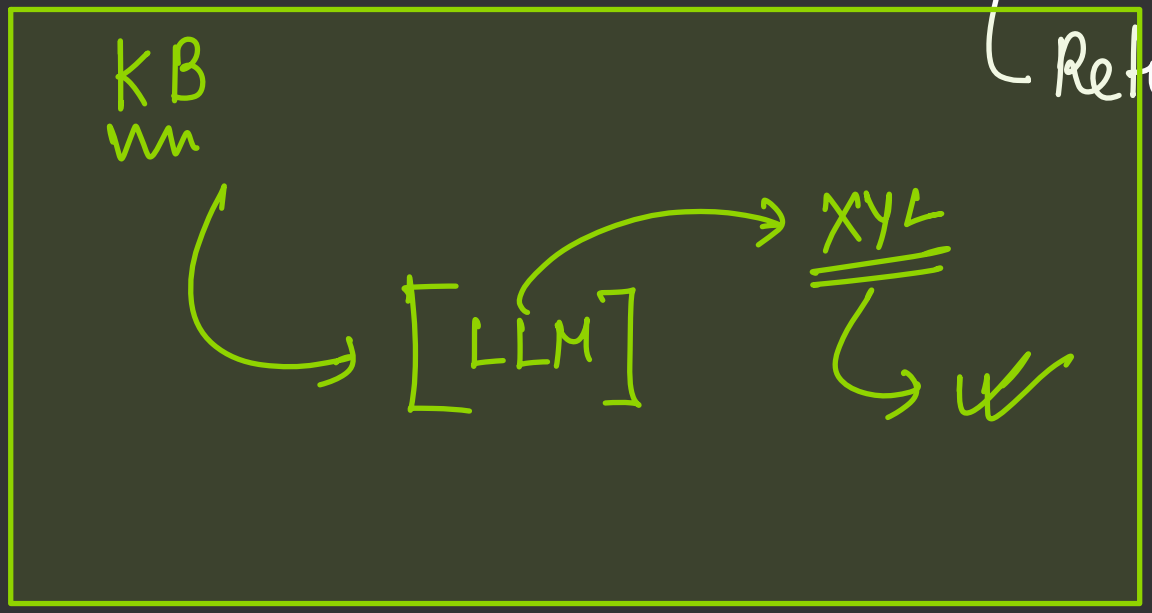

Analyzed
COT
ZSP
FSP
Output generation
Cap token



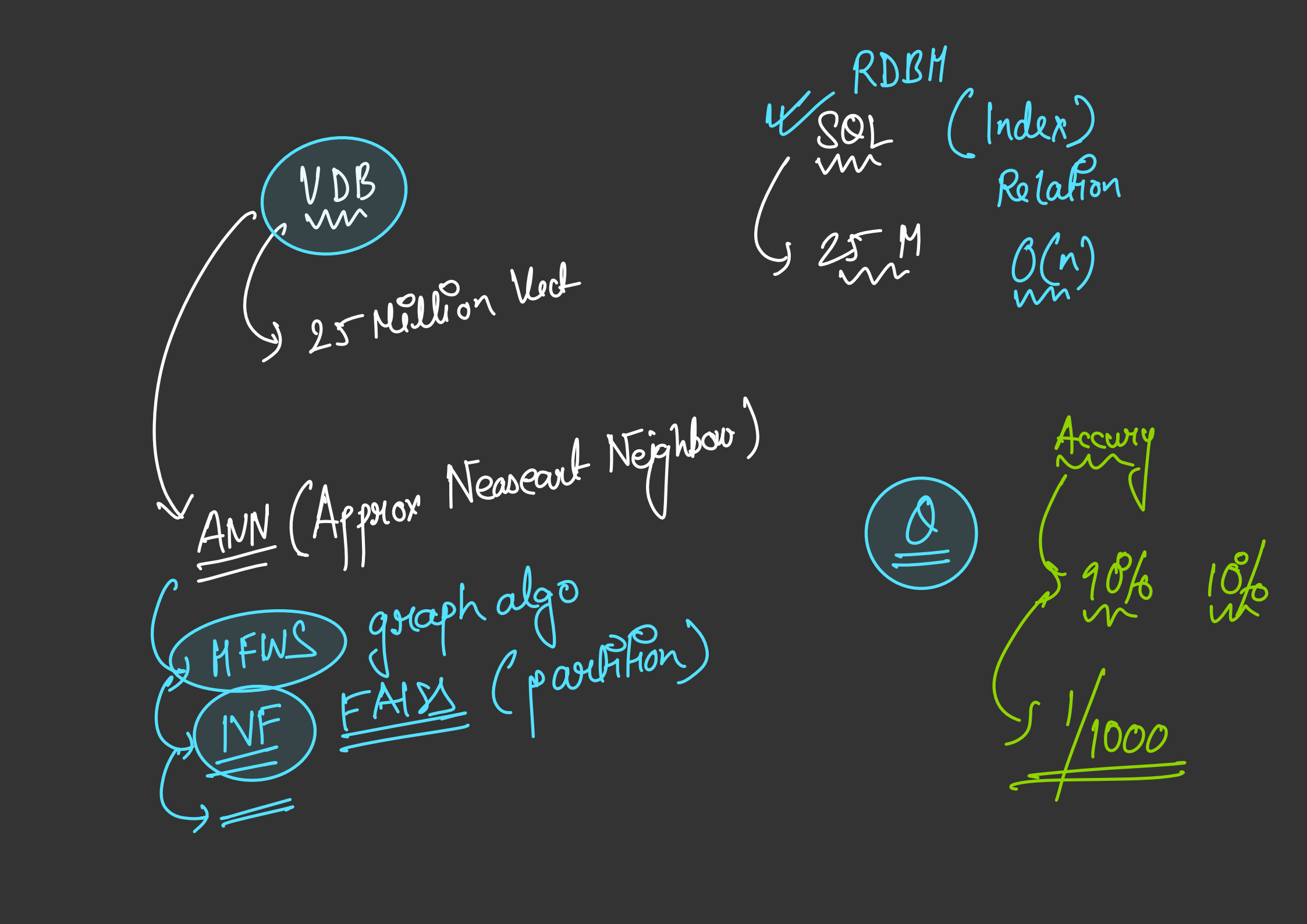
prompt

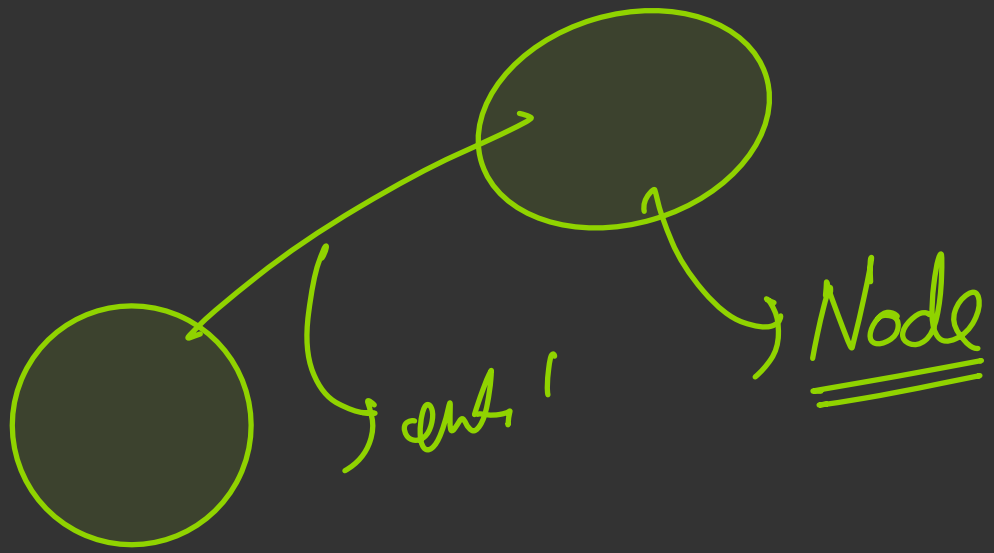
ReAct

- Define goal
- Analyse tools
- Return → Yes / No



RAG


RDBM
SQL
(Index)
Relation
O(n)
25 M
VDB
25 Million Vect
ANN (Approx Neaseart Nejghbou)
HFWS
IVF
graph algo
FAISS (partition)
Accury
90%
10%
1/1000



KB